

Linea Mixed Upgrade Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Linea on Linea Mixed Upgrade. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated March 27, 2026, was delivered to Linea, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Linea Mixed Upgrade code provided by Linea. The findings consist of 1 Critical & 1 Low severity issues with the remainder being informational.

Summary

Project Name	Linea Mixed Upgrade
Repository	linea-monorepo
Commit	78f9d1fa3ccc...
Fix Commit	c374a2c9f3a6...
Audit Timeline	Jan 5th - Jan 9th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	1	1	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	1	1	0
Informational	6	4	2
Gas Optimizations	0	0	0
Total	8	6	2

6 Audit Scope

This was an incremental audit of the following mix of changes:

- dynamic chain variables
- contracts folder restructuring
- data modularization
- transient storage

The audit scope was limited to the following files.

Changes merged from the deleted branch `feat/996-dynamic-chain-variables`:

```
contracts/src/rollup/LineaRollupBase.sol
contracts/src/rollup/ZkEvmV2.sol
contracts/src/verifiers/PlonkVerifierForDataAggregation.sol
```

Changes merged from the deleted branch `contract-folder-restructure`:

```
contracts/src/bridging/token/TokenBridge.sol
contracts/src/bridging/token/TokenBridgeBase.sol
contracts/src/security/reentrancy/TransientStorageReentrancyGuardUpgradeable.sol
contracts/src/messaging/l1/v1/L1MessageManagerV1.sol
contracts/src/messaging/l1/L1MessageService.sol
contracts/src/messaging/l2/v1/L2MessageServiceV1.sol
contracts/src/messaging/l2/L2MessageService.sol
contracts/src/messaging/l2/L2MessageServiceBase.sol
contracts/src/security/pausing/PauseManager.sol
```

Data modularization changes related to PR 1743:

```
contracts/src/messaging/l1/v1/ClaimMessageV1.sol
contracts/src/rollup/LineaRollup.sol
contracts/src/rollup/LineaRollupBase.sol
contracts/src/rollup/LivenessRecovery.sol
contracts/src/rollup/Validium.sol
contracts/src/rollup/dataAvailability/CalldataBlobAcceptor.sol
contracts/src/rollup/dataAvailability/Eip4844BlobAcceptor.sol
contracts/src/rollup/dataAvailability/LocalShnarfProvider.sol
contracts/src/rollup/dataAvailability/ShnarfDataAcceptor.sol
contracts/src/rollup/dataAvailability/ShnarfDataAcceptorBase.sol
```

7 Deployment Bytecode Verification

The audited functionality, including mitigations, was merged to the `main` branch in PR2007 and commit 5074237. After an additional audit by another provider, the resulting code was deployed on-chain to:

- `LineaRollup` (Ethereum): 0xE68697690E8ff196A6aBB3E1385156D87Df85332
- L1 `TokenBridge` (Ethereum): 0xF0e003F0dE2d583Ae28FA8cBF66aa096CdAce3ff
- `L2MessageService` (Linea): 0x9976fD7edDb78156a002DE74c9158E884702273d
- L2 `TokenBridge` (Linea): 0x4a496167F187A97379e763f693A499cE1182848b

At commit 698d8830c, Cyfrin verified that the locally compiled bytecode matched both the expected bytecode in the repository and the deployed bytecode on-chain, using the following reproducible process.

First, prepare the repository:

```
gh repo clone Consensys/linea-monorepo
cd linea-monorepo
git checkout 698d8830c359b698ceee510f0d2beedebac9f7fe
pnpm install --ignore-scripts
cd node_modules/.pnpm/c-kzg@4.1.0/node_modules/c-kzg && npx node-gyp rebuild
cd contracts
npx hardhat test
```

Second, diff the locally compiled bytecode against the expected bytecode:

```
diff build/src/rollup/LineaRollup.sol/LineaRollup.json \
  deployments/bytecode/2026-02-17/LineaRollup.json
diff build/src/messaging/l2/L2MessageService.sol/L2MessageService.json \
  deployments/bytecode/2026-02-17/L2MessageService.json
diff build/src/bridging/token/TokenBridge.sol/TokenBridge.json \
  deployments/bytecode/2026-02-17/TokenBridge.json
```

Finally, for each deployed contract, fetch the on-chain bytecode with `eth_getCode` (Ethereum mainnet for `LineaRollup` and the L1 `TokenBridge`; Linea mainnet, public RPC `https://rpc.linea.build`, for `L2MessageService`

and the L2 `TokenBridge`) and compare it with the `deployedBytecode` of the local build artifact, after stripping the CBOR metadata and ignoring the positions where the local bytecode holds zero bytes for immutable values. The comparison succeeded for all four contracts.

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

Linea is a Type 2 zkEVM L2 rollup with full EVM equivalence which aims to provide the "Ethereum experience" at scale and has been previously audited by Cyfrin.

10 Attestation

Cyfrin attests that it performed a security review of the Linea Mixed Upgrade code provided by Linea.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-03-27-cyfrin-linea-mixed-upgrade-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Linea has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=99a433b86003511e5b0b75064fe53c55335a75f0a64461a8fcda58164be54bff
F=2026-03-27-cyfrin-linea-mixed-upgrade-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.